

Watching someone print, without you

Milestone 6. The only milestone that proves anything, and it has never been attempted. Fill this in as it happens - not afterwards.

The one rule: do not help.

Not a hint, not a nudge, not "try tapping that". The moment you help, the run stops being evidence - you have replaced the thing being measured with yourself. If they ask you something, write the question down and say:

"Pretend I'm not here. Do whatever you'd do on your own."

The urge to help will be strong, because you built it and you can see the button they are missing. **That urge is the data.** If they are genuinely stuck and distressed, stop, and record it as a hard stop. A hard stop is the most valuable result this can produce.

The goal

Not to find out whether the app works. To find out **where the person stops**. Those are different questions and only the second has ever produced a fix here.

Who

One person who has never used this, does not 3D print, and was not in any conversation about it. A partner or friend is fine; a colleague who has heard you talk about it is not - they will already know what "plate" means, and that is the thing being tested.

One person is enough for the first run. Do not line up five. The first will hit so much that running the others before fixing anything wastes four people.

Before they arrive

- ☐ Printer on, filament loaded, already paired to Bambu Handy.
- ☐ MakerWorld account already signed in on the iPad's Safari. (Testing signup is a different, longer test - and it is Bambu's flow, not ours.)
- ☐ Bambu Handy installed and signed in to the same account.
- ☐ A real model already saved in the iPad's Files app - something from MakerWorld **with more than one loose part**, so the split button is reachable. Not a test cube.
- ☐ The app open at the /local page. Nothing else open.
- ☐ iPad auto-lock set long. A screen sleeping mid-task is your bug, not theirs.
- ☐ Screen recording running, if they consent. You will miss things live.
- ☐ This sheet, a pen, and a clock.

What you say

Once, then stop talking:

"This is a thing for printing a 3D model on my printer. There's a file called _____ in Files. Please make it print. Think out loud if you can."

Do not say: plate, part, orient, mesh, slice, container, upload, MakerWorld, or Handy. If you name a step, you have taught them the step.

Marks

- P** a pause over about ten seconds, no action

W a wrong turn - had to undo or go back
- Q** they asked you something (write it down)

X hard stop

Write down **verbatim** anything they call something by a different name than the app does - "the thingy", "the printy button". That is the app's vocabulary being wrong, and it is the cheapest fix you will ever get.

Time	What they did / said	Mark

Where the evidence says they will stall

Predictions, written in advance so you notice them live rather than reconstruct them after. **A prediction that does not fire is as useful as one that does** - it means attention is going to the wrong place.

- ☐ **1. Getting started at all**
Does the Files picker show the model, or grey it out? An accept attribute on the file input made iPads refuse every file once. It is gone and must not come back.

- ☐ **2. Three files at the end**
New and never watched. The output is the model, a picture, and a how-to page. Do they save all three? Can they tell which is which? **Do they ever open the how-to page?** If they ignore it, it is not instructions - it is a file nobody opens, and the premise needs rethinking.

- ☐ **3. The photo, at MakerWorld**
The known worst step. MakerWorld refuses our render and demands a real photo. The how-to page warns about this. Do they read it, believe it, or grind against the form first? Predicted: a long pause and a Q.

- ☐ **4. Private**
If they miss it the model is public. Note it, do not correct it during the run, fix it yourself afterwards.

- ☐ **5. Finding it in Bambu Handy**
Two routes; the fallback exists because the short one is easy to miss. Which do they take? Does the wording still match what Handy shows today? That copy was verified 2026-08-23 and Bambu ships updates.

- ☐ **6. Colour**
New, deliberate, and ours. Colour is only a picture - the printer uses whatever filament is loaded. If they pick blue and expect blue out of the printer, that is a real expectation mismatch. The panel says so in words. Find out whether words are enough.

- ☐ **7. Size**
Does "Keep its shape" mean anything to them? Do they notice when the model is bigger than the bed? The bar turning red is the only signal.

- ☐ **8. Split**
Predicted: they never touch it, because nothing tells them to. If they hit the bed limit and do not find Split, that finding is worth more than the button.

What counts as done

Done = the printer is printing. Not "they got to Handy", not "the file uploaded". Filament moving.

Time to printing, from your one sentence

Q count - every question they had to ask

Hard stop? Where?

There is no target number. This run establishes the number; later runs are measured against it.

Afterwards - ask in this order, write before discussing

"Talk me through what you thought was happening, at each point."

"Was there a moment you thought it had gone wrong?"

"If you had to do it again tomorrow, what would you dread?"

Then do not fix anything for a day

The urge after watching is to patch the specific button they missed. Most of what you saw will be one or two underlying problems wearing several costumes, and you can only see that once the sting wears off. Write the findings down first; decide second. They go in docs/HANDOFF.md, to the same standard as everything else there: what was observed, not what it means.

What this is not

Not a demo - if you find yourself presenting, stop. Not a usability score - one person is a flashlight, not a sample. Not a test of MakerWorld or Bambu Handy, which will produce failures you cannot fix - record those anyway, because the delivery loop rests on them.